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Unimolecular decomposition in a spherically symmetric potential

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Unimolecular decompositions are considered where the interaction between the separating fragments can be described by a function solely of their radial separation. It is shown how previous treatments, confined to idealized models, can be generalized. Kinetic energy distributions and rate constants may be calculated accordingly when the angular momentum is low. Applications to evaporation from van der Waals molecules are noted. A limit to the interaction parameters is then found where these properties become "thermodynamic," despite the absence of angular momentum. This result is used to obtain a general form for the effects of angular momentum on a rate constant. It reduces to traditional ideas in the hard-sphere limit.

I. INTRODUCTION

Among the earliest examples of final-state distributions from a unimolecular reaction were measurements of the kinetic energies with which the fragments separate. They were also the sort of distributions that invites modeling. The earliest experiments typically involved molecular ions and so it seemed plausible that these distributions might be understood in terms of an attractive ion-induced dipole interaction overlaid with a centrifugal potential. This appeared in fact to be the case.¹

Chesnavich *et al.*² pointed out in 1981, however, that this did not really make much sense. It had become clear by that time that the absolute rate constants for these dissociations were often not compatible with the spherically symmetric potentials implicit in this modeling. Low steric factors pointed instead to oriented moieties and tight transition states. The earlier successes would seem to have been at least in part illusory.

With the onset of measurements of kinetic energy release in evaporation from van der Waals molecules, the use of long-range potentials has acquired a new lease on life. In this case, however, it is a hard-sphere potential (together with the ubiquitous centrifugal component) which first comes to mind as the appropriate model to use.³ Nevertheless, ions continue to be the subject of most experimental studies. Consider, e.g., the reaction



It has been investigated experimentally in several quarters⁴⁻⁶ and clearly calls for interpretation with a superposition of the hard-sphere and ion-induced dipole models.

We present here the background for such an interpretation. It will be assumed that, as the two fragments separate, their interaction is described adequately by

$$V = -\xi/r^4: r > b, \quad (2a)$$

$$V = -\infty: r < b, \quad (2b)$$

where b is the hard-sphere collision radius. At first, only parent species of low angular momentum will be considered. Relations will be presented for calculating the kinetic energy distribution which attends any such potential. It will then be shown how, when the agreement with experiment is satisfactory, information pertaining to the thermodynamics of the evaporation can be deduced. The magnitudes of the absolute rate constants will also be addressed and a general theorem will be obtained. It will be seen to augment our understanding of the role of angular momentum in chemical kinetics. In particular, it will lead us to an expression for rate constants as a function of angular momentum as well as of the energy.

II. THEORY

When the rotational temperature is much less than that of the other degrees of freedom, the (unnormalized) kinetic energy distribution of the fragments, after separating in a spherically symmetric potential, is given by^{1,3}

$$p(\epsilon) \sim \int_0^{E-E_0-\epsilon} \rho_V(x) (E-E_0-\epsilon-x)^{s-1} dx \\ - \int_0^{E-E_0-\epsilon-BL_{\max}^2} \rho_V(x) \\ \times (E-E_0-\epsilon-BL_{\max}^2-x)^{s-1} dx, \quad (3)$$

where ϵ is the kinetic energy, E is the total energy of the isolated molecule, E_0 is the energy threshold for the reaction, $\rho_V(x)$ is a density of the vibrational states of the products, $2s$ is equal to one less than the number of final rotational degrees of freedom, and where the rotational constant B is equal to $\hbar^2/2(I_1+I_2)$, where I_1 and I_2 are the moments of inertia of the product fragments. $L_{\max}$ the maximum value of the orbiting angular momentum quantum number is a function of the kinetic energy. For the hard-sphere model, it is given by

$$(\hbar L)_{\max}^2 = 2\mu\epsilon b^2, \quad (4)$$

where μ is the reduced mass, while for the Langevin model, it is given instead by

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$$(\hbar L)_{\max}^2 = 4\mu(\xi\epsilon)^{1/2}, \quad (5)$$

where ξ is the constant appearing in Eq. (2). The analogous expression for a composite model may be obtained without difficulty. To whatever potential seems appropriate, one adds the centrifugal term $(\hbar L)^2/2\mu r^2$. The maximum in this net quantity is found as a function of the radial separation and defines a potential barrier. It will block all kinetic energies less than its magnitude. Alternatively, it defines a maximum value of the orbiting angular momentum permitting the emergence of a given kinetic energy.

For a composite potential such as in Eq. (2), the outcome of this prescription is quite simple. At low kinetic energies, the centrifugal barriers will occur at radial separations greater than the hard-sphere collision radius and so will be unaffected by its existence. Equation (5) will thus still be correct. The magnitude of this barrier can never exceed the value it obtains at the hard-sphere radius, however. At large kinetic energies, one may therefore write

$$(\hbar L)_{\max}^2 = 2\mu b^2(\epsilon + \xi/b^4). \quad (6)$$

The transition between the two criteria given in Eqs. (5) and (6) occurs smoothly at the energy where they are equivalent and thus when ϵ equals ξ/b^4 .

One can generalize this procedure to any combination of a centrifugal and an attractive potential, although it is likely that additional nuances will only rarely be called for. Let us therefore turn to the practical matter of using these criteria to evaluate Eq. (3). This might be done in several ways. We use a method which is both simple and instructive.

The first term on the right-hand side of that equation is the phase-space term. If there were no centrifugal barriers, it alone would be present. The kinetic energy distribution would then be a monotonically decreasing function and very nearly that of a two-dimensional Maxwellian gas. Let us denote with $k_{\text{ps}}(E)$ the phase-space rate constant that would result from an integration of Eq. (3) over just this term. It also pertains only to systems with low angular momentum and so may be written as a function of energy only. Consider then the hard-sphere criterion in Eq. (4). An integration over both terms in Eq. (3) yields for $k_{\text{hs}}(E)$ the hard-sphere rate constant

$$k_{\text{hs}}(E) = k_{\text{ps}}(E) [\mu b^2 / (\mu b^2 + I_1 + I_2)]. \quad (7)$$

We observe similarly that $\bar{\epsilon}_{\text{hs}}$ the average kinetic energy in the hard-sphere approximation is related to its analog in the phase-space limit by

$$\bar{\epsilon}_{\text{hs}} = \bar{\epsilon}_{\text{ps}} [2(I_1 + I_2) + \mu b^2] / (I_1 + I_2 + \mu b^2). \quad (8)$$

It will therefore be assumed that the kinetic energy distribution (3) may be written in the (also unnormalized) form

$$p(\epsilon) \sim \exp(-\epsilon/k_B T^\ddagger) [1 - \exp(-BL_{\max}^2/k_B T^\ddagger)] \quad (9)$$

and so, upon insertion of the appropriate forms for $L_{\max}^2$, may be evaluated. We note that this form preserves the properties enumerated in Eqs. (7) and (8).

One then observes that the average kinetic energy in the phase space limit is given by the relation

$$\bar{\epsilon}_{\text{ps}} = k_B T^\ddagger. \quad (10)$$

There are no centrifugal barriers in this limit and so $T^\ddagger$ can be thought of as the temperature of the transition state, but consider the temperature defined by the general expression

$$E = \Delta E_{\text{act}}(T^\ddagger) + \bar{E}_p(T^\ddagger) - k_B T^\ddagger, \quad (11)$$

where $\Delta E_{\text{act}}(T^\ddagger)$ is an Arrhenius activation energy and where $\bar{E}_p(T^\ddagger)$ is the thermal energy of the parent species, each evaluated at the particular angular momentum under consideration. It has been shown⁷ that in the case of chemistry, this temperature can also be thought of as that of the transition state. We therefore identify the parameter which appears in Eq. (9) with that in Eq. (11).

Examination of Eq. (8) then shows that the average kinetic energy is bounded (in the hard-sphere model) by

$$k_B T^\ddagger \leq \bar{\epsilon} \leq 2k_B T^\ddagger. \quad (12)$$

The first bound is the phase-space limit, while the second might be called the thermodynamic limit. It results when the rotational temperature, instead of being negligible, is equal to that of the other degrees of freedom. Equation (8) shows that it will also be attained when $(I_1 + I_2) \gg \mu b^2$, no matter how low the angular momentum is.

The presence of an attractive component in the potential will lower centrifugal barriers, but the limits given by Eq. (12) will not be changed. Suppose, however, that one wants to measure this temperature more precisely. A plausible procedure would be to assign values to b , I_1 , I_2 , and ξ , the several hard-sphere and Langevin parameters. The temperature in Eq. (9) might then be varied until a best fit to an experimental distribution is obtained.

If the energy of the ions which produced the distribution were known, this temperature could be entered into Eq. (11) to determine an activation energy. In typical experiments, however, these energies are not known. Nevertheless, one may proceed as follows: The parameter $T^\ddagger$ has another important interpretation. It is the temperature of an infinite heat bath which would produce the observed kinetic energy distribution. It is linked by the expression⁸

$$T^\ddagger = T_b [1 - \Delta E_a / (2Ck_B T_b) + (\Delta E_a / Ck_B T_b)^2 / 12 \cdots \cdots] \quad (13)$$

to T_b , the isokinetic temperature, the setting of an infinite heat bath where the parent ions would dissociate with the same rate constant as those which produced the distribution. This temperature is linked in turn, in the special case of heavy-particle evaporation, to the activation energy by the semiempirical relation³

$$\Delta E_{\text{act}} \simeq 23.5 (\pm 1) k_B T_b \quad (14)$$

for species which evaporate with rate constants roughly equal to 10^5 (s)^{-1} . The heat capacity C which appears in

Eq. (13) is (in units of k_B) one less than that of the parent species at the isokinetic temperature. If it is estimated, one can then transform a temperature measurement into one of an activation energy.

Approximate versions of this procedure have been used before^{9,10} and the number of instances is encouraging where agreement has been found with heats of evaporation obtained by other means. Let us therefore consider the errors which can arise and how they might be minimized.

The uncertainty in the Gspann parameter, appearing in Eq. (14), is about 5%, although its effect will be less when comparing relative activation energies in a series of closely related evaporations. This uncertainty might also be reduced by using *ad hoc* information to calculate a Gspann parameter explicitly. Nevertheless, the range of evaporations that Eq. (14) pertains to is striking.

Preliminary calculations also show a remarkable insensitivity of the extracted temperature to details of the assumed potential parameters and moments of inertia. This can be understood as follows: for the hard-sphere model, one can write

$$\bar{\epsilon}^2 - \bar{\epsilon}^2 = (k_B T^\ddagger)^2 + (\bar{\epsilon} - k_B T^\ddagger)^2, \quad (15)$$

and so a measurement of the first two moments of a kinetic energy distribution would suffice to determine a temperature. There would be no need to guess at additional parameters. At one time⁸ it was therefore hoped that one could use a relation such as Eq. (15) to obtain a temperature directly. But any experimental measurement is subject to error. While the average of a quantity may be measured adequately, the second moment of a distribution will almost always be overestimated. Yet an examination of Eq. (15) will show that the extracted temperature will be quite sensitive to an error in that moment.

Fortunately, we have seen above that it is no longer necessary to depend on highly idealized potentials in order to make predictions. An investment of some attention in the details of the assumed potential and to the pertinent moments of inertia should therefore remove the model-making stage as a source of error. One can then adjust the temperature until the first moment of the distribution is reproduced, with the second moment serving as a check on the process. Later we will mention a data-smoothing procedure which might further enhance the accuracy of the extracted temperature.

III. THE THERMODYNAMIC LIMIT

Explicit examples of temperature extraction will be given elsewhere. Let us focus here instead on additional generalities which pertain to decomposition in a spherically symmetric potential. We begin by using Eqs. (7) and (8) to write

$$\bar{\epsilon}/k_B T^\ddagger + k_{\text{hs}}(E)/k_{\text{ps}}(E) = 2 \quad (16)$$

for the hard-sphere model. Thus as the parameters which characterize the model are varied, so that the average ki-

netic energy moves out to its thermodynamic limit, the rate constant becomes much less than the phase-space rate constant. One might then ask whether there will also be something "thermodynamic" about its magnitude in that same limit of the parameters. It will turn out that this is the case. Furthermore, the result is not limited to the hard-sphere model, but is quite general. It will thus illuminate the broader matter of angular momentum in chemical kinetics.

We must first define more carefully the limit we shall be investigating. Equation (9) contains the term $\exp(-BL_{\text{max}}^2/k_B T^\ddagger)$. It can be expanded to yield

$$\exp(-BL_{\text{max}}^2/k_B T^\ddagger) = 1 - BL_{\text{max}}^2/k_B T^\ddagger + \dots \quad (17)$$

When this expansion is truncated after just these two terms, the hard-sphere and other analytically solvable models yield average kinetic energies which agree with thermodynamic arguments. We shall therefore investigate the rate constants which ensue when the same truncation is made.

To obtain a general description of this limit, however, we must revert to the earlier expression of Eq. (3). In its framework, the limit under investigation occurs when that expression may be written as

$$p(\epsilon) \sim BL_{\text{max}}^2 \left[d \int_0^{E-E_0-\epsilon} \rho_v(x) \times (E-E_0-\epsilon-x)^{s-1} dx/dE \right]. \quad (18)$$

Let us denote by $k'(E)$ the limiting value of the rate constant which results upon integrating over this relation. Averaging it in turn over a Boltzmann distribution of energies will then give us the desired rate constant. It may be obtained explicitly by recalling that a phase-space rate constant can be written, in the limit of vanishing angular momentum, as

$$k_{\text{ps}}(E) = [h\rho_v^0(E)]^{-1} \int_0^{E-E_0} \rho_v(x) (E-x)^s dx/\beta, \quad (19)$$

where $\rho_v^0(E)$ is the density of vibrational levels of the parent species and β is a known function of the moments of inertia of the reaction products.¹¹ We may therefore write $k'(E)$ as

$$k'(E) = [h\rho_v^0(E)]^{-1} (sB/\beta) \int_0^{E-E_0} d\epsilon L_{\text{max}}^2 \times \left[d \int_0^{E-E_0-\epsilon} \rho_v(x) \times (E-E_0-\epsilon-x)^{s-1} dx/dE \right]. \quad (20)$$

We then observe that the integral on the right-hand side of this relation can be identified as the sum over states at the energy $E-E_0$ of a system whose partition function, at some arbitrary temperature, may be written as

$$Q = Q_{\text{vib}} \Gamma(s) (kT)^{s-1} \int \exp(-\epsilon/kT) L_{\text{max}}^2 d\epsilon/k_B T, \quad (21)$$

where Q_{vib} is a vibrational partition function for the product species. If we integrate $k'(E)$ over a Boltzman distribution of energies, we will therefore get

$$k'(T) = \exp(-E_0/k_B T) (k_B T/h) [B\Gamma(s+1) (kT)^{s-1}/\beta] \times (Q_{\text{vib}}/Q_{\text{vib}}^0) \int \exp(-\epsilon/kT) L_{\text{max}}^2 d\epsilon/k_B T, \quad (22)$$

where Q_{vib}^0 is the vibrational partition function of the parent species.

Recall that this result is correct only for species of low angular momentum and then only in the limit defined by Eq. (18). A thermodynamic rate constant on the other hand is subject to neither of these constraints. It may be obtained from microscopic reversibility and is given by

$$k_{\text{thermo}}(T) = K_{\text{eq}} k_{\text{assoc}}, \quad (23)$$

where K_{eq} is the dissociation equilibrium constant and k_{assoc} is the bimolecular rate constant for the association step. When this rate is determined solely by centrifugal barriers in a spherically symmetric potential, we may write

$$k_{\text{thermo}}(T) = \exp(-E_0/k_B T) (kT/h) (Q_{\text{vib}}/Q_{\text{vib}}^0) \times (Q_{\text{rot}}/Q_{\text{rot}}^0) \times \int \exp(-\epsilon/kT) L_{\text{max}}^2 d\epsilon/k_B T, \quad (24)$$

where Q_{rot} and Q_{rot}^0 are rotational partition functions for the product fragments and the parent molecule, respectively.

Upon comparing this result with Eq. (22), and upon inserting the explicit forms of the function β , we find

$$k'(T) = k_{\text{thermo}}(T) [I_0/(I_1 + I_2)]^{3/2}, \quad (25)$$

where I_0 is the geometric-mean moment of inertia of the parent molecule. Thus, aside from the factor containing some moments of inertia, the two rate constants agree. We therefore have a general theorem. When the parameters which describe a reaction rate are varied so that the kinetic energy distribution goes over to the thermodynamic limit, a thermally averaged rate constant very nearly does also. In this limit, the absence of angular momentum is of essentially no consequence.

It might seem that the factor in Eq. (25) containing the moments of inertia constitutes a blemish on this result. Nevertheless, it will be clear below how that factor must be there in order to achieve an even more striking correspondence. We note for now that the relation provides a bridge between molecules with and without angular momentum. Let us see how it can be exploited.

IV. RABINOVITCH FACTORS

The preceding discussion concerned rate constants in a particular limit, specifically when the expansion in Eq. (17) is sufficient. But for an arbitrarily chosen molecule this will usually not be the case, nor are we at liberty to vary its parameters so as to make the truncation permissible. A rate constant at a low angular momentum will then typically not be linked so simply to a thermodynamic quantity. Nevertheless, we may assume that any difference between the two is related to the error which is made in truncating that expansion prematurely. Let us therefore seek an expression for $k_0(T)$, the rate constant which results when no truncation is made, but with the angular momentum still assumed to be small.

The ratio $k_{\text{thermo}}(T)/k_0(T)$ is also of interest. It might be called the Rabinovitch factor. When transition states are highly localized, this factor is typically found to be given by a ratio $(I^\ddagger/I_0)^{3/2}$, where $I^\ddagger$ is a moment of inertia of the transition state. Waage and Rabinovitch¹² also considered the rate constant ratio for "loose" transition states such as those we are considering here. They used an extension of the Rice-Gershinowitz model, which treats the dissociating species as a diatomic molecule. The final orbiting angular momentum quantum number is then of necessity equal to J_0 , that of the parent species. Not surprisingly, the resulting expression differs from what we will obtain.

In order to obtain $k_0(T)$, we must average $k(E)$ over a Boltzman distribution of energies. This task is not done as easily as was the averaging of $k'(E)$, its truncated form. We proceed by noting that the two rate constants are related by

$$k(E) = k'(E) \int \exp(-\epsilon/k_B T^\ddagger) \times [1 - \exp(-BL_{\text{max}}^2/k_B T^\ddagger)] d\epsilon / \int \exp(-\epsilon/k_B T^\ddagger) (BL_{\text{max}}^2/k_B T^\ddagger) d\epsilon, \quad (26)$$

where, as noted before, E and $T^\ddagger$ are connected via Eq. (11). One now observes that the parameters which one might vary, in order to pass to the thermodynamic limit, are examples of *intramolecular* parameters. They do not appear in the density of states of the parent species. For such a parameter, say α , and when E and $T^\ddagger$ are connected as above, we may use the theorem¹³

$$\partial \ln k(T^\ddagger)/\partial \alpha = \partial \ln k(E)/\partial \alpha \quad (27)$$

to relate the effect of a variation $\partial \alpha$ on a thermal rate constant with its effect on a microcanonical rate constant. Recall now that $k_0(T)$ and $k'(T)$ are thermal rate constants, each pertaining to species of low angular momentum, but evaluated, respectively, without and with the truncation described in Eq. (17). Because of Eqs. (26) and (27), we may then write

$$k_0(T)/k'(T) = \int \exp(-\epsilon/k_B T) [1 - \exp(-BL_{\max}^2/k_B T)] d\epsilon / \int \exp(-\epsilon/k_B T) (BL_{\max}^2/k_B T) d\epsilon. \quad (28)$$

This relation pertains in principle only at the temperature $T^\ddagger$ connected with E by Eq. (11), but since that energy was chosen arbitrarily, the result may be considered quite general. Then, on comparing it with Eq. (25), we obtain

$$k_{\text{thermo}}(T)/k_0(T) = [(I_1 + I_2)/I_0]^{3/2} \int \exp(-\epsilon/k_B T) (BL_{\max}^2/k_B T) \times (d\epsilon/k_B T) / \int \exp(-\epsilon/k_B T) \times [1 - \exp(-BL_{\max}^2/k_B T)] d\epsilon/k_B T \quad (29)$$

for the Rabinovitch factor.

This expression can be readily evaluated for an arbitrary potential. Let us use the hard-sphere model as a test to see if it makes sense. Consider two such spheres with moments of inertia I_1 and I_2 , and let $r^\ddagger$ be the distance between their respective centers of mass in the transition state for their mutual separation. This transition state will be a prolate symmetric top with moments of inertia equal to $I_1 + I_2$ and $I_1 + I_2 + \mu r^{\ddagger 2}$. The standard result for tight transition states then gives

$$k_{\text{thermo}}(T)/k_0(T) = [(I_1 + I_2 + \mu r^{\ddagger 2})/I_0] [(I_1 + I_2)/I_0]^{1/2} \quad (30)$$

for the Rabinovitch factor. We recognize now that the hard-sphere model is only a special case of this more general model. It is particular in that the two spheres are free to rotate around their respective centers, and in that we have been using the symbol b for $r^\ddagger$. With that understanding, an evaluation of Eq. (29) should yield Eq. (30). It does.

The principal difference between the present result and that of Waage and Rabinovitch lies in the integral in the denominator on the right-hand side of Eq. (29). It is effectively set equal to unity in the earlier work, and so the two expressions can agree only in the phase-space limit. It is not difficult to see explicitly why this should be the case. A diatomic model uses $\mu r^{\ddagger 2}$ as the moment of inertia, whereas in a more general treatment, the contributions I_1 and I_2 must also be included. Their neglect is justified only when $\mu r^{\ddagger 2}$ is much greater than $I_1 + I_2$. That is just the criterion which delineates the phase-space limit.

We may therefore have some confidence in Eq. (29). It has important consequences. Our earlier expressions, e.g., give us

$$k(E) = k_{\text{ps}}(E) \int \exp(-\epsilon/k_B T^\ddagger) \times [1 - \exp(-BL_{\max}^2/k_B T^\ddagger)] d\epsilon/k_B T^\ddagger \quad (31)$$

for a rate constant, but this result is valid only for species of low angular momentum. The energy here is then entirely in the vibrations. Suppose now that the molecule is set spinning with the vibrational energy gradually lowered such that the total energy remains constant. It is shown elsewhere¹⁴ that in these circumstances one may use the relation

$$\partial \ln k(E, J_0) / \partial J_0^2 = B_0/k_B T - B^\ddagger/k_B T^\ddagger, \quad (32)$$

where B_0 is the rotational constant of the parent molecule, $T^\ddagger$ is the temperature defined by Eq. (11), and T is similarly a temperature given by

$$E = \bar{E}_p(T, J_0) - k_B T, \quad (33)$$

where $\bar{E}_p(T, J_0)$ is again a canonical energy of the parent molecule. The remaining parameter $B^\ddagger$ may be obtained by identifying the ratio $(B_0/B^\ddagger)^{3/2}$ with the Rabinovitch factor for the reaction evaluated at the temperature $T^\ddagger$.

Nuances involved in integrating Eq. (32) are discussed elsewhere¹⁴ and present little in the way of difficulties. Except when the angular momentum is very large, as with collision complexes, it will usually suffice to write

$$\ln[k(E, J_0)/k(E)] = [(B_0/k_B T) - (B^\ddagger/k_B T^\ddagger)] J_0^2. \quad (34)$$

Thus a specification of a spherical potential permits one to calculate kinetic energy distributions with Eq. (9), rate constants with Eq. (31), and finally angular momentum effects via Eq. (32).

We lack only a simple formula for kinetic energy distributions at some arbitrarily high angular momentum. But when the rotational temperature is roughly that of the other degrees of freedom, even this task is straightforward. One may then use the thermodynamic limit, as represented by Eq. (17), to calculate the distribution.

V. DISCUSSION

A key stage in arriving at this happy state was the derivation of Eq. (29). It constitutes an easily surveyable expression for Rabinovitch factors and permits us to see more clearly the role of angular momentum in this class of unimolecular reactions. In particular, it would appear that these factors are given by what can be interpreted as a ratio $(I^\ddagger/I_0)^{3/2}$, just as with more conventional transition states. This conclusion was in turn arrived at following the identification of a limit. The existence of this limit, in which the absence of angular momentum is of no consequence, is also interesting. It had been noted earlier³ that thermodynamically grounded results such as the Richardson equation for thermionic emission could be obtained as a limit to the hard-sphere model. We now see that this correspondence will occur quite generally.

The abstract possibility of varying parameters invoked above actually exists in the case of evaporation from van der Waals molecules. This can be seen most easily by resorting again to the hard-sphere model. By treating each fragment as a uniformly filled sphere, one can show that the ratio $\mu b^2/(I_1 + I_2)$ will vary between the limits

$$2.5 \gg \mu b^2 / (I_1 + I_2) \gg 0 \quad (35)$$

as the process changes from a symmetric fissioning to a highly asymmetric evaporation. These two limits can be identified with evaporation from a dimer and a very large aggregate, respectively. The first bound closely approaches the phase space limit, while the second is the thermodynamic limit.

How might this transition be observed? The bounds described in Eq. (12) offer one possibility and are attractive in addition because their validity is not restricted to the hard-sphere model. A plot of the ratio $\bar{\epsilon}/k_B T^\ddagger$ as a function of aggregate size might therefore be expected to change more or less monotonically. As noted earlier, however, the temperature will usually not be known *a priori*. The relative variance in the kinetic energy distribution, on the other hand, is an entirely operational construct. The hard-sphere model predicts the bounds

$$1 \geq (\bar{\epsilon}^2 - \epsilon^2) / \epsilon^2 \geq 0.5 \quad (36)$$

for this quantity, and so it might be used to chart the transition.

Such a plot could serve a second role. The hard-sphere model also predicts that the relative variance will be independent of the temperature $k_B T^\ddagger$ for a given set of the parameters μ , b , I_1 , and I_2 . With a more complicated potential, this independence is lost, at least in principle. Model calculations show, however, that any such dependence is weak. It is in fact just this circumstance which permits the scaling of kinetic energy distributions.¹⁵ A plot of the relative variance should thus be a monotonic function of cluster size and so deviations from this behavior may be assumed to be experimental in origin. A smoothing of the data would then permit a more precise assignment of the average energies and hence of the underlying temperatures.

Finally, we should address the question whether relations derived for decompositions in a spherically symmetric potential pertain only to evaporation. As a corollary, we may ask why they have worked so well for reactions with low steric factors.

The statistical adiabatic channel model is often used to discuss such reactions. In its simplest version, the potential along the optimal path of approach is characterized with a Morse potential and its familiar parameter β . A second parameter α is used to describe the evolution of the degrees of freedom of the separated fragments into those of the parent molecule with which they correlate. With these assumptions, the course of a reaction in a highly asymmetric potential can be written formally in terms of a single radial variable. It is then not difficult to calculate rate constants

and product distributions. Let us consider one example. It has been found that the ratio $\alpha/\beta = 0.5$ suffices to rationalize a large number of experimental measurements.¹⁶ Troe also finds¹⁷ that this value implies that the ratio of $k_{\text{acm}}(E)$, an adiabatic channel rate constant, to $k_{\text{ps}}(E)$, a phase-space rate constant, is independent of energy. This result recalls Eq. (7) for a hard-sphere rate constant. Closer inspection then shows that Troe's result implies that Eqs. (12) and (15) are also valid, as is the analog of Eq. (16)

$$\bar{\epsilon}/k_B T^\ddagger + k_{\text{acm}}(E)/k_{\text{ps}}(E) = 2. \quad (37)$$

A kinetic energy distribution calculated with a plausible model for a "tight" transition state can thus mimic a standard result of an isotropic potential.

It is therefore not surprising that relations developed in an idealized context have seemed to work more generally. But this mimicry should not be overinterpreted. In pursuing the adiabatic channel model further, one finds for example that values of α/β less than one-half can lead to kinetic energy distributions which contravene the limits prescribed by Eq. (12). One must therefore distinguish between potential surfaces of pseudo- and those of genuinely spherical symmetry. The relations presented here pertain strictly only to the latter.

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